

Weakening interaction suppresses spontaneous symmetry breaking in two-channel asymmetric exclusion processes

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This paper has studied spontaneous symmetry breaking (SSB) phenomenon in two types of two-channel asymmetric simple exclusion processes (ASEPs). One common feature of the two systems is that interactions for each species of particle happen at only one site, and the system reduces to two independent ASEPs when interaction vanishes. It is shown that with the weakening of interaction, the SSB is suppressed. More interestingly, the SSB disappears before the interaction is eliminated. Our work thus indicates that local interaction has to be strong enough to produce SSB. The mean-field analysis has been carried out, and the results are consistent with the simulation ones.

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I. INTRODUCTION

The analysis of driven diffusive systems is a central issue of nonequilibrium statistical mechanics. These systems show a variety of generic nonequilibrium effects, in particular, in low dimensions. A very prominent example of such a system is the asymmetric exclusion processes (ASEPs), which are discrete nonequilibrium models that describe the stochastic dynamics of multiparticle transport along one-dimensional lattices. Each lattice site can be either empty or occupied by a single particle. Particles interact only through hardcore exclusion potential. ASEPs were introduced originally as a model to explain the processes of transcription (creation of the messenger RNA) and translation (creation of proteins) in 1968 [1] and now have become an important tool to investigate various nonequilibrium phenomena in chemistry, physics, and biology. They have been applied successfully to understand polymer dynamics in dense media [2], diffusion through membrane chains [3], gel electrophoresis [4], dynamics of motor proteins moving along rigid filaments [5], kinetics of synthesis of proteins [6,7], and dynamics of traffic flow [8,9].

Many nonequilibrium behaviors, such as boundary-induced phase transition, phase separation, dynamical scaling, and spontaneous symmetry breaking (SSB), have been observed in ASEPs [10–20]. The first model that exhibits SSB was a model with open boundaries, and it is known as a “bridge model” [10]. In the model, two species of particles move in the opposite directions. It was shown that two phases with broken symmetry could exist, despite the update rules are symmetric with respect to the two species. While it is argued by Monte Carlo simulations that one of the phases (i.e., asymmetric low-density/low-density phase) may not exist in the thermodynamic limit [10,16], mean-field analysis shows that it could exist in a very small region.

Pronina and Kolomeisky [19] pointed out that in the single-channel two-species ASEP, particles of different species interact with each other at every site of the lattice. Thus it is not clear how the interactions between these particles localized in specific parts of the system will affect the symmetry breaking. They have proposed to study a two-channel ASEP with narrow entrances. They found surprisingly that local interaction at the entrances strongly influences macroscopic particle dynamics of the system.

Yuan *et al.* have studied another two-channel ASEP in which the two channels intersect at a center site [20]. It has been found that SSB also occurs when removal rate β is smaller than a critical value β_c and injection rate $\alpha > \beta$.

This paper focuses on the SSB phenomenon in the two-channel ASEPs in the two references mentioned above, in which interactions for each species of particle happen at only one site (boundary site in Ref. [19] and center site in Ref. [20]). It is demonstrated that when the interaction is weakened (but not eliminated), the SSB disappears. This indicates that local interaction has to be strong enough to produce SSB.

The paper is organized as follows. We present the two generalized models in the next section. In Sec. III, both the analytical investigations by using the mean-field approximation and the Monte Carlo simulations have been carried out. Finally, the conclusion is given in Sec. IV.

II. MODELS

The two models studied in Refs. [19,20] have been generalized as follows. In both models the random update rules are adopted.

In model A, the system consists of two parallel one-dimensional L lattices with two species of particles moving in different lanes in the opposite directions [Fig. 1(a)]. The hopping between the lanes is not allowed. In the bulk, a particle hops to the next site with rate 1 provided the target site is empty. At the exit site, a particle is removed with rate β . At the entrance site, a particle is injected with rate α provided this site and exit site on the other lane are empty,

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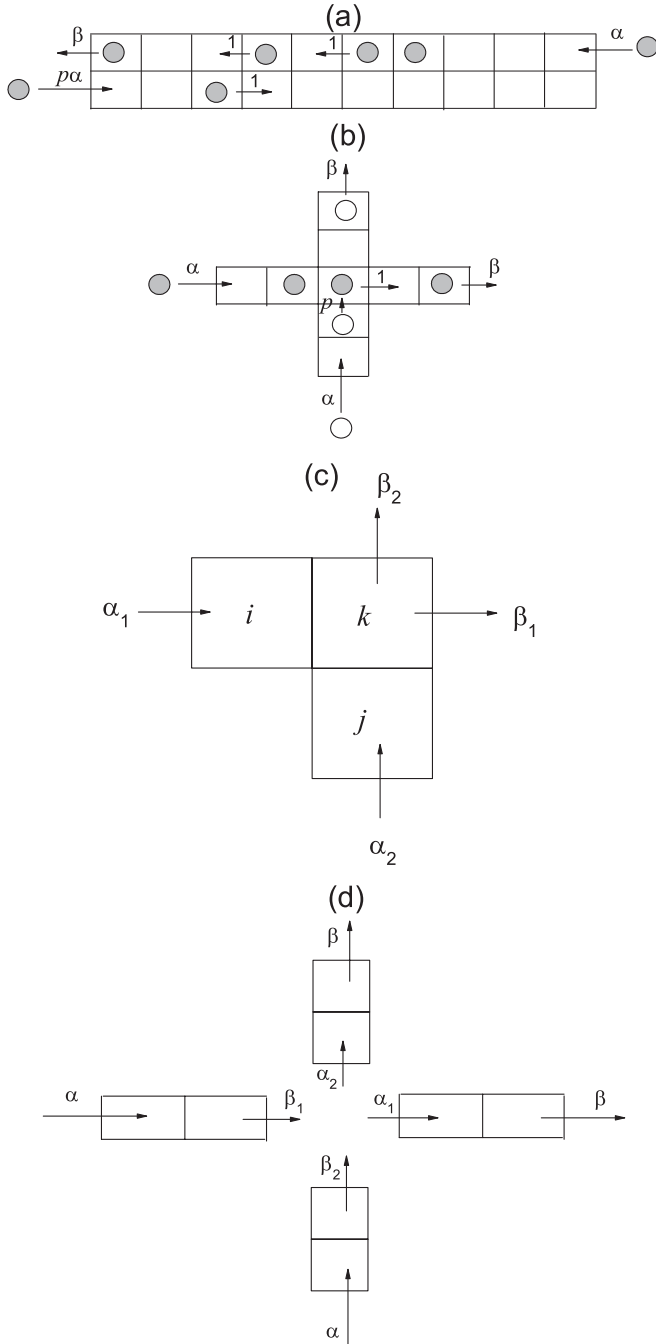


FIG. 1. Sketch of (a) model A, (b) model B. The number indicates the hopping rate. (c) The sketch of the three sites and effective injection rates and removal rates used in mean-field analysis. (d) The sketch of effective injection rates and removal rates used in simple mean-field analysis.

and with rate $p\alpha$ provided this site is empty and the exit site on the other lane are occupied. Here p controls the strength of interaction at the boundaries. In the special case $p = 0$, the model reduces to the one studied in Ref. [19]. In another special case $p = 1$, the model reduces two independent single-channel ASEPs.

In model B, the system consists of two intersected one-dimensional L lattices with two species of particles moving in different lanes [Fig. 1(b)]. In the intersected site, it is allowed to hold two particles of different species. In the bulk (except the two sites upstream of the intersected site), a particle hops to the next site with rate 1 provided the target site is empty. Note that in the intersected sites, two particles of different species could move simultaneously. In the sites upstream of the intersected site, a particle hops to the intersected site with rate 1 provided the target site is empty and with rate p provided the target site is occupied by a particle of different species. At the exit site, a particle is removed with rate β . At the entrance site, a particle is injected with rate α provided this site is empty. Similarly, here p controls the strength of interaction at the intersected site. In the special case $p = 0$, the model reduces to the one studied in Ref. [20]. In another special case $p = 1$, the model reduces two independent single-channel ASEPs.

III. ANALYTICAL AND SIMULATION RESULTS

A. Model A

Figure 2 shows the phase diagram of model A at different values of p , obtained from Monte Carlo simulations. When p is small, there exist five phases: two asymmetric ones [i.e., asymmetric low-density-low-density phase (LD-LD) and asymmetric high-density/low-density phase (HD/LD)] and three symmetric ones [i.e., symmetric high-density phase (HD), symmetric maximum-current phase (MC), and symmetric low-density phase (LD)]; see Fig. 2(a).

With the increase of p , the two asymmetric phases shrink. When $p = 0.4$, the LD-LD phase becomes too narrow to be distinguished from simulations [see Fig. 2(b)]. The HD phase and the MC phase expand. The LD phase shrinks in the vertical direction and expands in the horizontal direction. When p exceeds a critical value $p_c \approx 0.6$, the two asymmetric phases disappear [Fig. 2(c)].

Mean-field analysis of the phase boundaries in model A could be carried out, following the methodology in Ref. [19], by denoting the effective entrance rates as α_1 and α_2 :

$$\begin{aligned}\alpha_1 &= \alpha(1 - m_1) + p\alpha m_1 = \alpha[1 - (1 - p)m_1], \\ \alpha_2 &= \alpha(1 - p_L) + p\alpha p_L = \alpha[1 - (1 - p)p_L].\end{aligned}\quad (1)$$

Here p_L and m_1 are density on site L of lane 1 and density on site 1 of lane 2, respectively. The mean-field calculation is straightforward to carry out, and details will not be presented here. The analytical results are listed as follows:

The asymmetric HD/LD is determined by

$$\frac{p\alpha}{1 - \alpha(1 - p)} < \beta < \frac{\alpha - p\alpha - p\alpha^2 + p^2\alpha^2}{1 + \alpha + \alpha^2 - p\alpha - 2p\alpha^2 + p^2\alpha^2}.\quad (2)$$

The symmetric LD phase is determined by

$$\alpha < \frac{2\beta}{4\beta - 1 + p}.\quad (3)$$

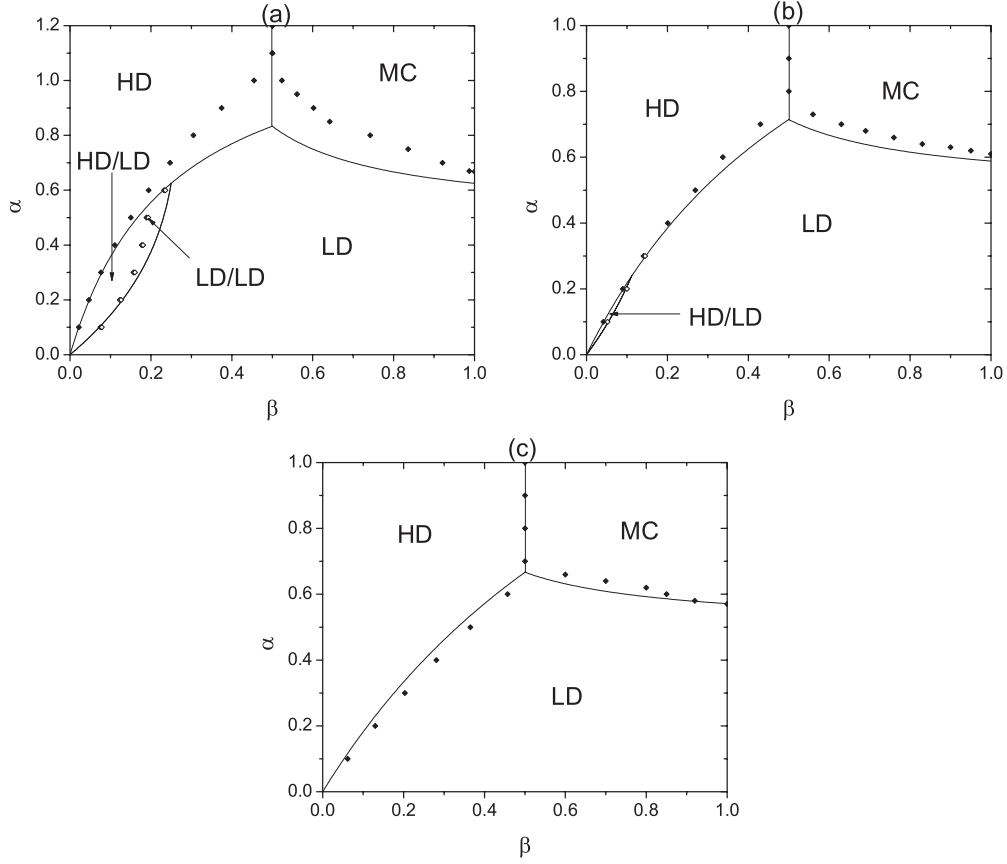


FIG. 2. Phase diagram of model A. (a) $p = 0.2$, (b) $p = 0.4$, (c) $p = 0.6$. The solid lines are from mean-field analysis, the scattered points are from Monte Carlo simulations. The LD-LD phase from the mean-field analysis is too narrow to be visible.

The existence of asymmetric LD-LD phase requires

$$\frac{\alpha - p\alpha - p\alpha^2 + p^2\alpha^2}{1 + \alpha + \alpha^2 - p\alpha - 2p\alpha^2 + p^2\alpha^2} < \beta < \frac{1}{6} [2\alpha - 4\alpha^2 - 2p\alpha + 4p\alpha^2 + \sqrt{(2\alpha - 4\alpha^2 - 2p\alpha + 4p\alpha^2)^2 + 12(\alpha^2 - 2p\alpha^2 + p^2\alpha^2)}]. \quad (4)$$

The symmetric HD-HD phase exists when

$$\beta < \min \left(\frac{p\alpha}{1 - \alpha(1 - p)}, \frac{1}{2} \right). \quad (5)$$

Finally, the symmetric MC/MC phase is determined by

$$\alpha > \frac{2\beta}{4\beta - 1 + p}, \quad \beta > 1/2. \quad (6)$$

We can also calculate the critical value of p at which the SSB disappears. This is achieved by considering the intersection point of the boundary between the symmetric HD phase and asymmetric HD/LD phase, with the two boundaries of the asymmetric LD-LD phase. The coordinates of the intersection point are calculated by setting

$$\frac{1}{6} [2\alpha - 4\alpha^2 - 2p\alpha + 4p\alpha^2 + \sqrt{(2\alpha - 4\alpha^2 - 2p\alpha + 4p\alpha^2)^2 + 12(\alpha^2 - 2p\alpha^2 + p^2\alpha^2)}] = \frac{p\alpha}{1 - \alpha(1 - p)}, \quad (7)$$

$$\frac{\alpha - p\alpha - p\alpha^2 + p^2\alpha^2}{1 + \alpha + \alpha^2 - p\alpha - 2p\alpha^2 + p^2\alpha^2} = \frac{p\alpha}{1 - \alpha(1 - p)}. \quad (8)$$

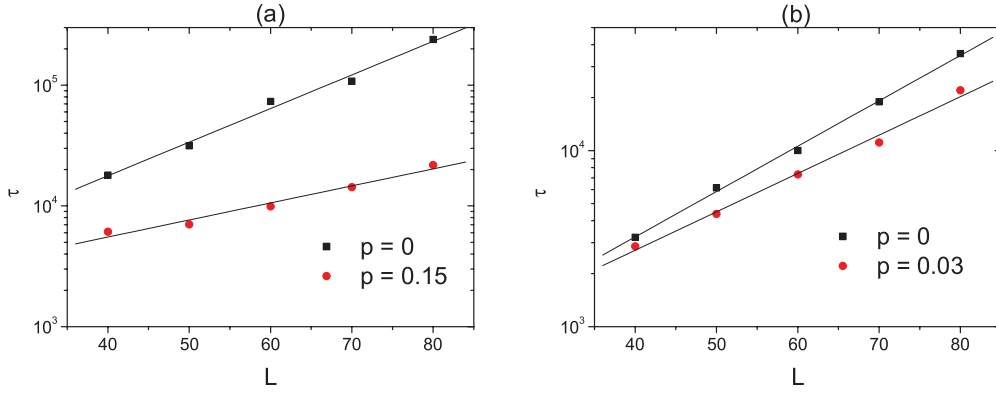


FIG. 3. (Color online) Dependence of characteristic flipping time scale on system size and interaction strength in (a) model A with parameters $\alpha = 0.5, \beta = 0.2$, (b) model B with parameters $\alpha = 0.7, \beta = 0.2$. The solid lines are guide for eyes.

For a general value of p , the analytical solutions of the two equations are difficult to obtain. [21] Fortunately, in the special case $p = 0.5$, the two equations could be simplified, and both solutions are $\alpha = \beta = 0$. Therefore, mean-field analysis predicts that when $p > p_c = 0.5$, the SSB disappears.

Figure 2 also shows the phase boundaries obtained from mean-field analysis. One can see that the results are consistent with the simulations. However, since correlations are ignored, there is deviation between simulation results and mean-field ones. With the increase of p , the correlations become weaker, thus the deviation shrinks.

We also study the dependence of characteristic flipping time scale τ on system size and interaction strength. As shown in Fig. 3(a), τ increases exponentially with system size, which indicates that the model does reproduce SSB. On the other hand, τ as well as growth rate of τ decreases with the weakening of interaction strength.

Finally we would like to mention that Gupta *et al.* have studied the robustness of SSB in a generalized bridge model [13]. In their model, exchange of the two species of particles could happen at the two boundaries: A westbound (an eastbound) particle could change into an eastbound (a westbound) particle with rate γ at the left (right) boundary. In the limit case $\gamma = 0$, their model reduces to the original bridge model. Gupta *et al.* found that when γ exceeds a critical value, the SSB disappears.

Although our work on model A reproduces similar results as Ref. [13], we would like to mention that there exists a qualitative difference. In Ref. [13] for each species of particle, the interactions happen at both entrance and exit boundaries, as well as at every site in the bulk. With the increase of γ , both the effective injection rate and effective removal rate change [see, e.g., Eq. (14) in Ref. [13,22]. Thus, the contributions of the two boundaries to the disappearance of SSB are not clear. In contrast, in our model, the interactions happen only at the entrance boundary. Therefore, the contribution of the entrance boundary is clearly identified. In particular, when SSB disappears, the generalized bridge model produces LD and MC phases but could not exhibit the HD phase [23]. This is different from our two models (see also model B below), in which the system reduces to two independent ASEPs when interaction disappears.

B. Model B

This subsection studies model B, and the phase diagram is shown in Fig. 4. As reported in Ref. [20], in the case of $p = 0$, there exist three phases [Fig. 4(a)]: two symmetric ones and one asymmetric phase. In a symmetric LL phase, the system is in low density [Fig. 5(a)]. In a symmetric HL phase, the system is in high density upstream of the intersection site and in low density downstream of the site [Fig. 5(b)]. In asymmetric phase, one channel is in high density, the other channel is in HL state [Fig. 5(c)].

With the increase of p , the LL phase expands, the HL phase shrinks. Moreover, a new symmetric HH phase appears [Fig. 4(b)], in which the system is in high density [Fig. 5(d)]. The HH phase expands with the increase of p and gradually invades the asymmetric phase. When p exceeds a critical value $p_c \approx 0.1$, the asymmetric phase disappears [Fig. 4(c)]. Figure 3(b) shows the characteristic flipping time scale τ in model B, which is similar as in model A.

Next we present mean-field analysis to investigate the asymmetric phase. We would like to mention that simple mean-field analysis fails to describe model B, as shown in Appendix A. Therefore, we consider the three sites as shown in Fig. 1(c).

Let $P_{\tau_1 \tau_2 \tau_3}$ denote the probability that site i in state τ_1 (0 means empty, 1 means occupied), site j in state τ_2 (0 means empty, 1 means occupied), site k in state τ_3 (0 means empty, 1 means occupied by an eastbound particle, 2 means occupied by a northbound particle, 3 means occupied by both an eastbound particle and a northbound particle); α_1 and α_2 denote effective injection rates into site i and j ; β_1 and β_2 denote effective removal rates of eastbound particle and northbound particle from site k ; J_1 and J_2 denote flow rates of eastbound particles and northbound particles; ρ_0 denotes probability that site k is occupied by northbound particle; and ρ_1, ρ_2 denote densities of sites i, j , respectively.

Now we can write the master equations of $P_{\tau_1 \tau_2 \tau_3}$. For instance,

$$dP_{000}/dt = -(\alpha_1 + \alpha_2)P_{000} + \beta_1 P_{001} + \beta_2 P_{002} + \beta_1 \beta_2 P_{003}. \quad (9)$$

Other master equations are shown in Appendix B.

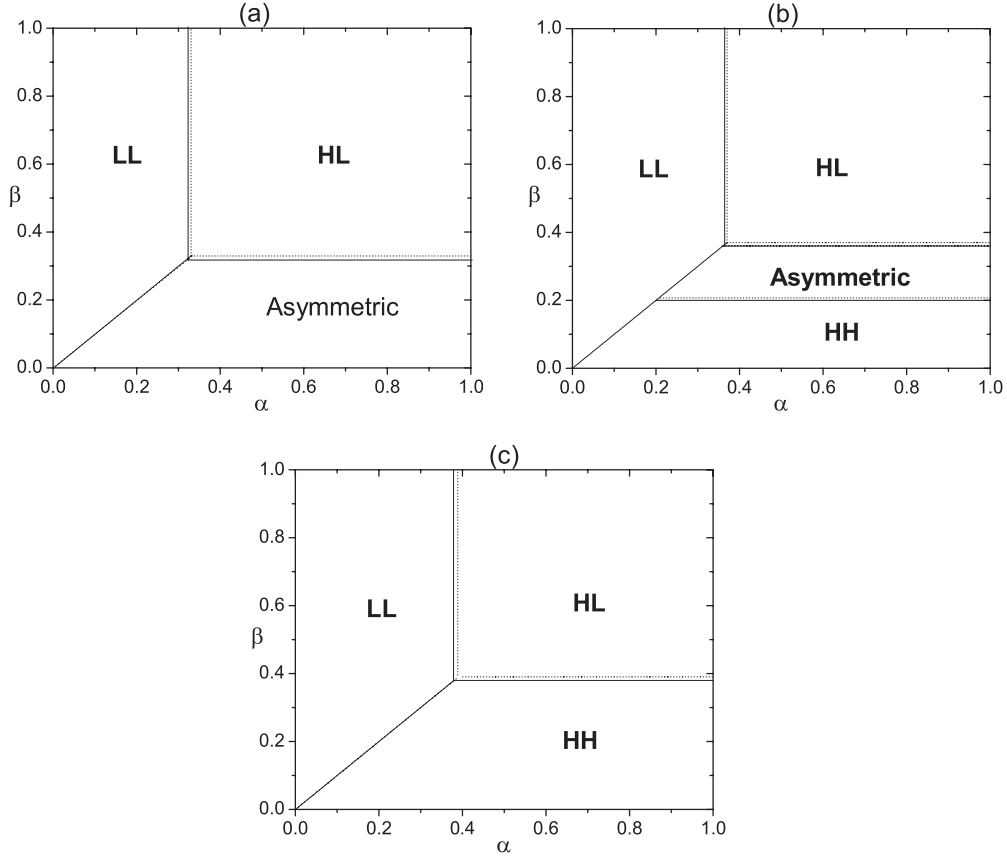


FIG. 4. The phase diagram of model B, obtained from Monte Carlo simulations (dashed lines) and mean-field analysis (solid lines). The parameter is (a) $p = 0$, (b) $p = 0.05$, (c) $p = 0.15$.

In the stationary state, we have $dP_{\tau_1\tau_2\tau_3}/dt = 0$. Therefore, we obtain 16 equations from Eqs. (9) and (B1)–(B15). However, only 15 of them are independent ones. The conservation of probability requires that

$$\begin{aligned} &P_{000} + P_{001} + P_{002} + P_{003} + P_{010} + P_{011} + P_{012} + P_{013} \\ &+ P_{100} + P_{101} + P_{102} + P_{103} + P_{110} + P_{111} \\ &+ P_{112} + P_{113} = 1 \end{aligned} \quad (10)$$

Furthermore, ρ_1 and ρ_2 are given by

$$\rho_1 = P_{100} + P_{101} + P_{102} + P_{103} + P_{110} + P_{111} + P_{112} + P_{113}, \quad (11)$$

$$\rho_2 = P_{010} + P_{011} + P_{012} + P_{013} + P_{110} + P_{111} + P_{112} + P_{113}. \quad (12)$$

On the other hand, by ignoring correlations, ρ_1 and J_1 , ρ_2 and J_2 satisfy

$$J_1 = \rho_1(1 - \rho_1), \quad (13)$$

$$J_2 = \rho_2(1 - \rho_2). \quad (14)$$

J_1 and J_2 can also be calculated by

$$J_1 = \alpha_1(1 - \rho_1), \quad (15)$$

$$J_2 = \alpha_2(1 - \rho_2). \quad (16)$$

In the asymmetric phase, one channel is in high density. Without loss of generality, we suppose that the eastbound

channel is in high density. As a result, J_1 is determined by $J_1 = \beta(1 - \beta)$. Substituting into Eq. (13), one obtains $\rho_1 > 0.5$ (the solution $\rho_1 < 0.5$ needs to be discarded). Then we can obtain $\alpha_1 = \rho_1$ via Eq. (15).

Finally, since downstream of the intersection site is in low density on the northbound lattice, one has

$$\rho_0 = J_2/\beta_2 = 1 - \rho_2. \quad (17)$$

Now we have 21 unknowns, including 16 probabilities $P(\tau_1\tau_2\tau_3)$, β_1 , β_2 , J_2 , α_2 , and ρ_2 . We also have 21 equations including Eqs. (9)–(12), (B1)–(B14), (14), (16), and (17). We solve the 21 equations to obtain ρ_1 and ρ_2 . We have plotted the difference between ρ_1 and ρ_2 versus β in Fig. 6. It can be seen that in the range $\beta_{c2} < \beta < \beta_{c1}$, the difference is positive. When β approaches β_{c1} or β_{c2} , the difference decreases. The difference disappears when $\beta = \beta_{c1}$ or $\beta = \beta_{c2}$. Therefore, the asymmetric phase exists in the range $\beta_{c2} < \beta < \beta_{c1}$ [24]. When $\beta > \beta_{c2}$, the system transits into the symmetric HL phase. Similarly, when $\beta < \beta_{c1}$, the asymmetric solution does not exist, and one could have only a symmetric solution, and the system is thus in the HH phase.

In Figs. 4 and 5, analytical results have been compared with Monte Carlo simulation ones. It can be seen that the analytical ones are in good agreement with the simulation ones. With the decrease of p , both β_{c1} and β_{c2} decrease. In the limit case $p = 0$, β_{c1} becomes zero. Thus, the HH phase cannot be observed. With the increase of p , both β_{c1} and β_{c2} increase.

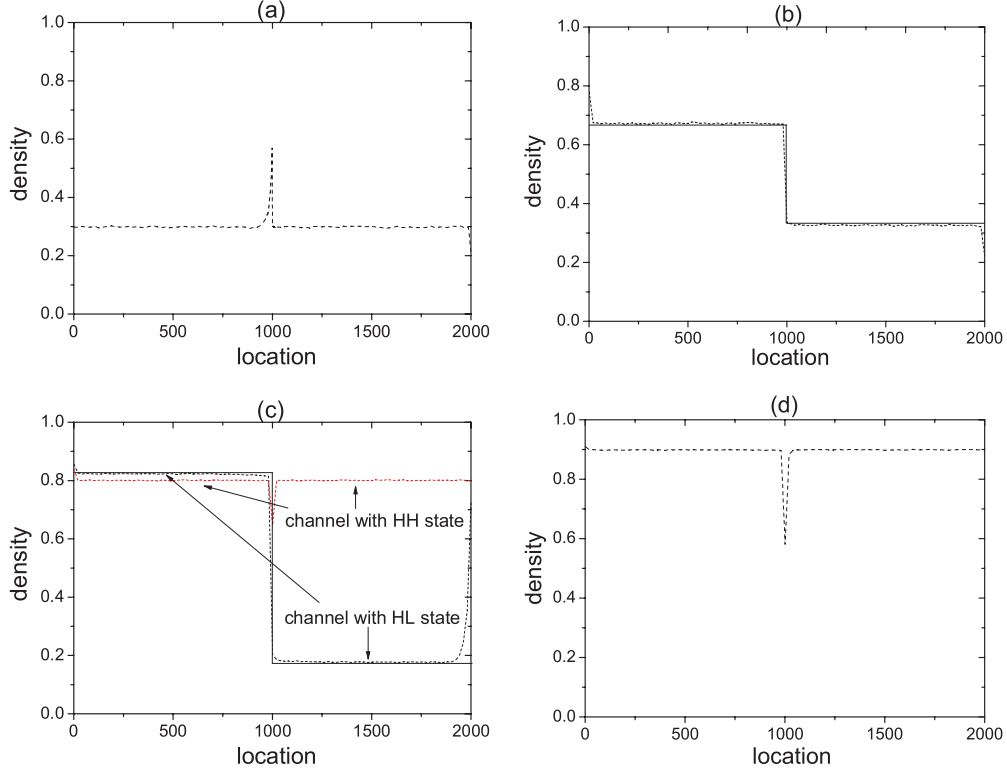


FIG. 5. (Color online) The density profiles obtained from Monte Carlo simulations (dashed lines) and mean-field analysis [i.e., Eqs. (11) and (12), solid lines] for the HL phase and the channel with HL state in asymmetric phase (solid lines). (a) The LL phase, with parameters $p = 0$, $\alpha = 0.3$, $\beta = 1$. (b) The HL phase, with parameters $p = 0$, $\alpha = 1$, $\beta = 1$. (c) The asymmetric phase, with parameters $p = 0$, $\alpha = 1$, $\beta = 0.2$. (d) The HH phase, with parameter $p = 0.05$, $\alpha = 1$, $\beta = 0.1$.

When $p = p'_c \approx 0.1$ (which is in good agreement with the simulation value $p_c \approx 0.1$), β_{c1} becomes equal to β_{c2} , and thus, the asymmetric phase disappears.

We can also calculate the boundary $\beta = \beta_c$ between the HH phase and the HL phase at $p > p'_c$. In the HL phase, due to symmetry, we have 14 unknowns, including 10 probabilities (excluding P_{010} , P_{012} , P_{013} , P_{112} , P_{002} , P_{011} , which equals P_{100} , P_{101} , P_{103} , P_{111} , P_{001} , P_{102} , respectively), β_2 , α_2 , J_2 , and ρ_2 . We also have 14 equations, including Eqs. (9), (B1), (B3), (B8)–(B13), (10), (12), (14), (16), and (17). We solve

the equations, and the boundary is determined by

$$\beta_c(1 - \beta_c) = J_2. \quad (18)$$

The analytical result is also in good agreement with the simulation; see Figs. 4 and 5(b). We also point out that in the range $p < p'_c$, β_c is identical to β_{c2} .

IV. CONCLUSION

To summarize, we have studied two models of two-channel ASEP, a parallel bidirectional two-channel ASEP with narrow entrance, and an ASEP consisting of two intersected lattices. In both systems, interactions for each species of particle exist only at one site, and the interaction could produce SSB. We have carried out mean-field analysis and numerical simulations for both models. In particular, we have explained why the symmetric HH phase could happen in model B when the parameter $p > 0$. Mean-field analysis and numerical simulations of both models show that with the weakening of interaction, the SSB is suppressed. Moreover, the SSB disappears before the interaction is eliminated. Our work thus indicates that local interaction has to be strong enough to produce SSB.

Finally we would like to mention that it is important to investigate how other one-site interaction or multisite interactions affect the system dynamics. Actually, we have already studied two-channel ASEP with narrow exits. It is found that the narrow exits lead to different results from the narrow entrances, in which no SSB has been observed.

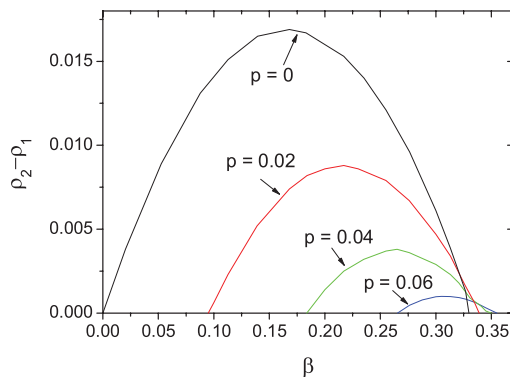


FIG. 6. (Color online) Dependence of $\rho_2 - \rho_1$ on β in the range $\beta_{c2} < \beta < \beta_{c1}$, obtained from mean-field analysis in model B.

Currently we are investigating the two-channel ASEPs with both narrow entrances and exits. In our future work, other multisite interactions will also be studied (e.g., narrow entrance/exit sections for two parallel ASEPs, two intersected ASEPs that share more than one site, relaxed exclusion situation for bridge model as mentioned in Sec. III A [23]).

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APPENDIX A

We denote the effective rates as α_1 , β_1 , α_2 , and β_2 [see Fig. 1(d)], which can be expressed as

$$\alpha_1 = \rho_{c1}, \quad (\text{A1})$$

$$\alpha_2 = \rho_{c2}, \quad (\text{A2})$$

$$\beta_1 = (1 - \rho_{c1})(1 - \rho_{c2}) + p(1 - \rho_{c1})\rho_{c2}, \quad (\text{A3})$$

$$\beta_2 = (1 - \rho_{c1})(1 - \rho_{c2}) + p(1 - \rho_{c2})\rho_{c1}. \quad (\text{A4})$$

Here ρ_{c1} and ρ_{c2} denote the density of particle of the two species at the intersection point, respectively.

In the symmetric HH phase, we have $\alpha_1 = \alpha_2$, $\beta_1 = \beta_2$, $\rho_{c1} = \rho_{c2}$. Due to flow rate conservation, we have

$$\beta_1 = \beta. \quad (\text{A5})$$

Therefore, from Eqs. (A3) and (A4), we have

$$(1 - \rho_{c1})(1 - \rho_{c1}) + p(1 - \rho_{c1})\rho_{c1} = \beta. \quad (\text{A6})$$

Thus, ρ_{c1} can be obtained

$$\rho_{c1} = \frac{2 - p - \sqrt{(p - 2)^2 - 4(1 - p)(1 - \beta)}}{2(1 - p)}. \quad (\text{A7})$$

Therefore, it is required that

$$(p - 2)^2 - 4(1 - p)(1 - \beta) > 0, \quad (\text{A8})$$

which yields

$$\beta < 1 - \frac{(p - 2)^2}{4(1 - p)}. \quad (\text{A9})$$

The right-hand side of the equation is always negative when $0 < p < 1$. As a result, mean-field analysis predicts that the symmetric HH phase could not exist.

APPENDIX B

This appendix gives other 15 master equations:

$$dP_{001}/dt = -(\alpha_1 + \alpha_2 + \beta_1)P_{001} + P_{100} + \beta_2(1 - \beta_1)P_{003}, \quad (\text{B1})$$

$$dP_{002}/dt = -(\alpha_1 + \alpha_2 + \beta_2)P_{002} + P_{010} + \beta_1(1 - \beta_2)P_{003}, \quad (\text{B2})$$

$$dP_{003}/dt = -(\alpha_1 + \alpha_2 + \beta_1(1 - \beta_2) + \beta_2(1 - \beta_1) + \beta_1\beta_2)P_{003} + (P_{102} + P_{011})p, \quad (\text{B3})$$

$$dP_{010}/dt = -(\alpha_1 + 1)P_{010} + \alpha_2P_{000} + \beta_1P_{011} + \beta_2P_{012} + \beta_1\beta_2P_{013}, \quad (\text{B4})$$

$$dP_{011}/dt = -(\alpha_1 + \beta_1 + p)P_{011} + P_{110} + \alpha_2P_{001} + \beta_2(1 - \beta_1)P_{013}, \quad (\text{B5})$$

$$dP_{012}/dt = -(\alpha_1 + \beta_2)P_{012} + \alpha_2P_{002} + \beta_1(1 - \beta_2)P_{013}, \quad (\text{B6})$$

$$dP_{013}/dt = -(\alpha_1 + \beta_1(1 - \beta_2) + \beta_2(1 - \beta_1) + \beta_1\beta_2)P_{013} + \alpha_2P_{003} + P_{112}p, \quad (\text{B7})$$

$$dP_{100}/dt = -(\alpha_2 + 1)P_{100} + \alpha_1P_{000} + \beta_1P_{101} + \beta_2P_{102} + \beta_1\beta_2P_{103}, \quad (\text{B8})$$

$$dP_{101}/dt = -(\alpha_2 + \beta_1)P_{101} + \alpha_1P_{001} + \beta_2(1 - \beta_1)P_{103}, \quad (\text{B9})$$

$$dP_{102}/dt = -(\alpha_2 + \beta_2 + p)P_{102} + P_{110} + \alpha_1P_{002} + \beta_1(1 - \beta_2)P_{103}, \quad (\text{B10})$$

$$dP_{103}/dt = -(\alpha_2 + \beta_1(1 - \beta_2) + \beta_2(1 - \beta_1) + \beta_1\beta_2)P_{103} + \alpha_1P_{003} + pP_{111}, \quad (\text{B11})$$

$$dP_{110}/dt = -2P_{110} + \alpha_1P_{010} + \alpha_2P_{100} + \beta_1P_{111} + \beta_2P_{112} + \beta_1\beta_2P_{113}, \quad (\text{B12})$$

$$dP_{111}/dt = -(\beta_1 + p)P_{111} + \alpha_1P_{011} + \alpha_2P_{101} + \beta_2(1 - \beta_1)P_{113}, \quad (\text{B13})$$

$$dP_{112}/dt = -(\beta_2 + p)P_{112} + \alpha_1P_{012} + \alpha_2P_{102} + \beta_1(1 - \beta_2)P_{113}, \quad (\text{B14})$$

$$dP_{113}/dt = -(\beta_1(1 - \beta_2) + \beta_2(1 - \beta_1) + \beta_1\beta_2)P_{113} + \alpha_1P_{013} + \alpha_2P_{103}. \quad (\text{B15})$$

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- [21] We would like to mention that numerical solutions indicate that when $p < p_c$, the solutions of the two equations are always identical; i.e., the three boundaries always intersect at the same point.
- [22] Actually it is hard to say whether interactions enhance or weaken with the increase of γ .
- [23] Since one site is allowed to hold only one particle in the bridge model, the sum of the densities of the two species particles cannot exceed one, which makes HD phase impossible. It is interesting to study the relaxed exclusion situation as in model B, i.e., if one site is allowed to hold two particles of different species.
- [24] We would like to mention that if we use $J_2 = J_1$ instead of Eq. (17), we could obtain a set of symmetric solution. We argue that since the asymmetric solution has a smaller flow rate on channel 2, the solution is stable and the symmetric one is unstable. Consequently, the asymmetric phase that corresponds to the asymmetric solution could exist, but the HH phase that corresponds to the symmetric solution could not be observed.